

ers. It is capable of driving both IGBTs and SiC MOSFETs and has a built-in 3.75kVrms isolator, which is higher than the 2.5kVrms isolator in the previous generation products. The IC



boasts superior common mode transient immunity at

150V/ns or higher, providing reliable communication and increased noise immunity while meeting the high voltages and fast switching speeds required in inverter systems. It can support power devices with a withstand voltage of up to 1200V, which makes it suitable to employ in cost-effective inverter systems. The gate driver MOSFET comes in a compact SOIC16 package. The IC offers multiple protection and fault detection functions, such as the on-chip active Miller clamp, soft turn-off, overcurrent (DESAT) protection, under-voltage lockout (UVLO), and fault feedback. The Renesas MOSFET driver can help increase EV adoption by allowing engineers to build cost-efficient inverters, thereby minimising environmental impacts.

Renesas

<https://www.renesas.com>

Radiation-tolerant LDO

MIC69303RT is Microchip's first commercial-off-the-shelf radiation-tolerant 3A low-dropout (LDO) voltage regulator that is suitable for objects in low-earth orbit (LEO). Operating from a single low-voltage supply of 1.65 to 5.5 volts, the device can supply output voltages as low as 0.5V at high currents, offering high precision and ultra-low dropout voltages of

500mV under extreme conditions. The MIC69303RT is a companion power source solution for Microchip's radiation-tolerant space-qualified micro-controllers, such as the SAM71Q21RT



and PolarFire FPGAs, including the RTPF500TLS.

Designed for harsh aerospace applications, the MIC69303RT is operational in temperature ranges from -55 to +125°C. It is offered in 8-pin and 10-pin packages with radiation tolerance up to 50Krad. The low noise of the output is critical to sensitive RF circuits, post regulation of switching power supplies, and industrial power applications.

Microchip

<https://www.microchip.com>

Image sensor for smartphones

OV50H is a 50MP image sensor from Omnivision that supports 12.5MP at 120 frames per second (fps) and high dynamic range (HDR) at 60fps. The image sensor will enable an aver-



age user to take noise-free, high-quality pictures even in low-light environments.

The OV50H is a 1.2-micron pixel image sensor in an 8.4mm optical format. The larger sensor size enables it to capture sharp images and allows for faster autofocus capability even in dark environments. The image sensor supports multiple HDR modes and high frame rates for smartphone rear-facing cameras. It is Omnivision's first sensor to feature horizontal/vertical quad-phase detection, which enables

users to capture stable images of fast-moving objects. It offers flagship-level low-light and autofocus performance, supports 12.5MP at 120fps and HDR at 60fps. The sensor can perform 4 cell binning, thus enabling a 12.5MP mode for video and preview with 2.4µm equivalent sensitivity for excellent low-light performance. Moreover, it supports up to 120fps with 4x low-light sensitivity and 8K video with on-chip remosaic. The sensor offers full 50MP Bayer output and 2x crop zoom functionality.

Omnivision

<https://www.ovt.com>

Gaming processor for smartphones

MediaTek's new Helio G36 processor will enable smartphone manufacturers to provide high-end smartphone specifications at an affordable price. The



processor will satisfy the growing demand for low-cost gaming smartphones in India. It offers a better camera, enhanced battery life, and greater

processing speed at an affordable cost. Helio G36 will enable the integration of better image sensors, which will allow for high-quality depth of field, and its higher processing power will allow the manufacturers to add their own AI camera features, such as AI beautification and enhanced precision. Helio G36 can run a 90Hz display and offer support for up to 50MP camera. It is built on a highly power-efficient TSMC 12nm production process, which ensures that the phone remains cool for an extended period. MediaTek has



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